

Accelerating USB 3.0 Implementations: Time is money

With the launch of chipsets incorporating USB 3.0 support, this latest USB interface standard is now becoming widely available on board level and system platforms. How long, though, will end-users in the industry have to wait before they can benefit from this technology in OEM products like vision systems? And how does the cost situation look?

INGRID EINSIEDLER

USB 3.0 is emerging as the new interface standard for fast peripherals. This need is evident in the field of appliances that can generate added value from this nominal 5Gbps performance boost beyond the maximum USB 2.0 performance (nominal 600 Mbps). With up to 2400 Mbps of usable data rate, it achieves a 7.5 times faster throughput than with USB 2.0 (320 Mbps).

Applications that profit from this performance boost are, for example, fast and high-resolution printers and scanners, or mass storage. Further examples are high-resolution rear-view cameras for all-round visibility in trucks, high-resolution test and measurement devices, and a vast number of vision applications like camera-supported pick-and-place (PnP), intelligent surveillance systems in the traffic technology, or digital microscopes, to name a few.

USB 3.0 focus market vision systems

In addition to Gigabit Ethernet, today in the vision area it is already becoming apparent that USB3 vision for close-range applications of up to five metres will evolve into the leading standard. The USB3 vision standard is entirely based on the USB 3.0 specification and additionally inserts GenICam—a generic programming interface that simplifies application porting and promotes interoperability of USB devices from different manu-



Fig. 1: The COM Express basic computer-on-module with type-6 pinout comes with 1x1.00GHz to 2x1.60GHz scalable Intel Celeron processor and HM65 PCH. With two optional USB 3.0 interfaces, it offers an especially cost-efficient USB 3.0 configuration, which has been available since the beginning of 2011 for early adopters, and is fitted with the NEC USB 3.0 Bridge

facturers. Customers benefit not only from the actual vision specifications but also from the advantages of USB 3.0 itself.

Advantages of USB 3.0

Apart from the immense performance boost—in comparison to USB 2.0—the USB 3.0 bus is backwards-compatible to USB 2.0 and 1.1. It further excels due to its simple cabling that provides power and data communication in one cable, which, in turn, serves to reduce installation work. Also, manufacturers are making their peripheral devices PnP-capable, so all you need for this is defined 'on the wire.' That results

in significantly less implementation work for the system integrator.

What's more, in future, USB 3.0 will be found on nearly all industrial computer platforms as USB 3.0 support will immediately be integrated into the chipsets—and presumably for a very long time to come. That saves a great amount of hardware development costs in comparison to standards like CameraLink, even though this standard—depending on the manufacturer—can process an 80-bit video track with up to 6.8 Gbps or 5400 Mbps, and continues to be supported via PCI- or PCIe-based framegrabber cards.

Consumer market pushes USB 3.0 cameras

If you observe the consumer camera market which is already switching to USB3.0, it becomes clear that the corresponding embedded camera technology can be used for close-range industrial applications. The main advantages therefore are reduced costs with added performance, more comfort and higher integration. Also, considering the fact that in video applications the processing of a multitude of tasks can be handed over to a GPU rather than to a framegrabber, in this area and in the long term, the PC ecosystem offers the